

1. Which gate outputs 1 only when all its inputs are 1?

- a) OR
- b) AND
- c) NAND
- d) XOR

Answer: b

2. Which gate is the complement of the AND gate?

- a) OR
- b) NAND
- c) NOR
- d) XOR

Answer: b

3. What is the binary representation of decimal 5?

- a) 101
- b) 110
- c) 100
- d) 011

Answer: a

4. How many bits are there in a nibble?

- a) 2
- b) 4
- c) 8
- d) 16

Answer: b

5. Which flip-flop has no invalid state?

- a) SR
- b) JK
- c) D
- d) T

Answer: b

6. A half adder has:

- a) 2 inputs and 1 output
- b) 2 inputs and 2 outputs
- c) 3 inputs and 2 outputs
- d) 1 input and 2 outputs

Answer: b

7. The hexadecimal digit F is equal to decimal:

- a) 10
- b) 15
- c) 16
- d) 12

Answer: b

8. A multiplexer selects:

- a) one of many inputs
- b) many of one input
- c) all inputs at once
- d) no input

Answer: a

9. Which of these is a sequential circuit?

- a) Adder
- b) Counter
- c) Decoder
- d) Multiplexer

Answer: b

10. According to De Morgan's law, $(A.B)'$ equals:

- a) $A' + B'$
- b) $A' + B$
- c) $A.B'$
- d) $A + B$

Answer: a

11. Explain the difference between combinational and sequential circuits.

(Write your answer in the space provided.)

12. Convert decimal 13 into binary and hexadecimal.

(Write your answer in the space provided.)